

CHEIRAP (■■■■■) — Complete Pitching Script & Jury Defense

National Innovation Challenge on AI & Digital Governance – Manipur (2026) | Host: DIT Manipur & MTIF

Focus Solution: Full Lifecycle Integrity	Stage 1: Pre-Award Tender Rigging Interception (GePNIC)	Total Time: 6 Minutes (Strict)
Submitted By: Team Lotux	Stage 2: Post-Award Works & Photo Forensics (PWD-04)	Format: 4m Pitch + 2m Q&A;

OFFICIAL INSTRUCTION FOR 6-MIN SLOT: Strict 4m presentation + 2m Q&A.; Demonstrate the complete lifecycle: how tender rigging pre-award leads to ghost construction post-award, and how CHEIRAP intercepts both.

MINUTE 1: The Connected Problem & Post-Mortem Audit Failure (0:00 – 1:00)

[STAGE & SCREEN ACTION — 0:00 – 1:00]: Stand tall, look at the jury. Display the CHEIRAP Surveillance Command Center showing ■1,874 Cr capex across 92 active tenders.

"Good morning, respected jury members. I am representing Team Lotux. Today, we are presenting **CHEIRAP**—the State Vigilance and Infrastructure Integrity Gateway.

Every year, the Government of Manipur spends hundreds of crores on capital works—schools, bridges, roads, and hospitals. Yet public infrastructure leakage happens in two linked stages:

- Stage 1 happens in the tender room:** collusive syndicates manipulate e-tenders before contracts are awarded.
- Stage 2 happens on the ground:** the same favored contractors claim public funds using fake or recycled construction photos.

Today, authorities discover this 2 to 3 years later through post-mortem CAG audits, after public money is already lost. We asked: Why wait for an autopsy? CHEIRAP protects public funds across the entire lifecycle: **pre-award tender interception** and **post-award construction verification**."

Target: 60s | Spoken Words: ~140 | Tone: Crisp, confident, simple plain English

MINUTE 2: Stage 1 — Pre-Award Tender Surveillance & Anti-Collusion (1:00 – 2:00)

[STAGE & SCREEN ACTION — 1:00 – 2:00]: Click on the 'Tender Surveillance Dashboard'. Open the Forensic Dossier and show the 4 Procurement Exploits.

"Stage 1 starts while bidding is live on GePNIC (manipur tenders.gov.in), **before bids are unlocked**. CHEIRAP continuously scans active tenders and intercepts the four major rigging exploits:

- 1. Window Squeeze:** Authorities publishing a tender with only 4 to 7 days before closing. CHEIRAP flags any window under the mandatory 21 days under GFR Rule 161.
- 2. Corrigendum Churn:** Uploading restrictive amendments 24 hours before bid closing without giving the mandatory 7-day CVC extension.
- 3. EMD Barriers:** Inflating Earnest Money Deposits to 5%–10% to choke outside MSMEs.
- 4. Cartel Clustering:** Using an Isolation Forest model to detect suspicious bid clusters.

When an exploit is detected, CHEIRAP generates a court-ready **Statutory Stay Order** citing Article 14 and Section 41(h) of the Specific Relief Act—halting tender manipulation before contracts are signed!"

Target: 60s | Spoken Words: ~150 | Tone: Crisp, confident, simple plain English

MINUTE 3: Stage 2 — PWD-04 Ground Construction Assurance (2:00 – 3:00)

[STAGE & SCREEN ACTION — 2:00 – 3:00]: Switch to 'Works Assurance'. Open Case A photo forensics modal (HUD viewfinder), then transition to Case B.

"Once a contract is awarded, CHEIRAP seamlessly transitions into Stage 2: **PWD-04 Construction Progress Monitoring**.

Contractors often commit fraud through **photo recycling**—submitting photos of older schools built years ago. A basic vision model sees bricks and gets fooled. CHEIRAP runs 4 rigorous engineering checks:

1. **Photo Forensics**: Calculates a 64-bit perceptual hash (pHash) and checks against 11,202 state projects.

2. **Geofence Boundary Check**: Validates camera EXIF GPS against registered survey polygons.

3. **CPWD Velocity Norms**: Enforces concrete curing physics—progress cannot jump 15% to 100% in 12 days.

4. **Measurement Book Parity**: Matches Running Account bills against verified physical engineer logs.

[Point to Screen] Look at **Case A**: A claimed Lamphelpat school photo matched an archived 2024 Bishnupur classroom with a **93.4% duplicate match**, and GPS drifted **9.4 kilometers** away. Running bills were frozen immediately! In contrast, **Case B** (Churachandpur College) showed 100% unique photos matching CPWD schedules, receiving automatic Green Passport clearance."

Target: 60s | Spoken Words: ~160 | Tone: Crisp, confident, simple plain English

MINUTE 4: Why CHEIRAP Wins & Closing Punchline (3:00 – 4:00)

[STAGE & SCREEN ACTION — 3:00 – 4:00]: Face the judges directly. Display the auto-generated PWD Form 44 Notice and Section 65B Certificate on screen.

"What makes our proposal uniquely different from any other hackathon project? Three critical factors:

1. **Administrative Due Process**: CHEIRAP is not an arbitrary AI black box. When anomalies are caught, it automatically generates a **PWD Form 44 Field Inspection Notice**, granting the Superintending Engineer a mandatory 7-day window to physically verify the site. It respects natural justice.

2. **Court-Ready Evidence**: Every alert, hash, and GPS record is cryptographically sealed under **Section 65B of the Bharatiya Sakshya Adhiniyam, 2023**. If challenged in the High Court of Manipur or Lokayukta, the evidence is legally admissible.

3. **Zero Workflow Disruption**: It plugs directly into existing state portals—GePNIC and Manipur Works MIS—without buying new hardware.

To conclude: Integrity in public money is not a post-mortem report. It is a real-time gatekeeper from the tender room to the construction ground. Thank you. We are ready for your questions."

Target: 60s | Spoken Words: ~155 | Tone: Crisp, confident, simple plain English

6-MINUTE BUZZER STRATEGY & PITCH TIMELINE:

• 0:00 – 1:00: Hook the jury on the 2 connected stages of capex leakage & post-mortem failure.

• 1:00 – 2:00: Show Pre-Award Tender Surveillance: the 4 exploits (window squeeze, GFR-161, stay order).

• 2:00 – 3:00: Show Post-Award Construction (PWD-04): pHash photo forensics, Case A vs Case B demo.

• 3:00 – 3:50: Deliver 3 Differentiators (Due Process, Sec 65B BSA Seals, Zero Disruption) + Punchline.

• 3:50 – 4:00: Smooth handover to the 2-minute Q&A; before the warning bell.

The 2-Minute Q&A; Session: Trilateral Jury Defense Cards

Direct, authoritative 30-second responses mapped to both Pre-Award Tenders and Post-Award Construction.

PRE-AWARD TENDER: LEGAL ENFORCEABILITY (35% WEIGHTAGE) — Anticipated Jury Question

Judge: "How can an AI system legally stay or pause a live e-tender before bids are opened?"

Core Strategy: Cite Article 14, GFR Rule 161, and Section 41(h) Specific Relief Act.

Your 30-Second Spoken Answer: "CHEIRAP does not replace the Vigilance Commissioner; it acts as a decision-support sentinel. Under GFR Rule 161 and CVC Directives, an open tender published with fewer than 21 days is void ab initio for violating Article 14. CHEIRAP flags this instantly and prepares a statutory hold recommendation under Section 41(h) of the Specific Relief Act. The Vigilance Commissioner digitally approves the notice, pausing bid unsealing before irreparable financial harm occurs."

POST-AWARD WORKS: TECHNICAL TRUST (PWD-04) (35% WEIGHTAGE) — Anticipated Jury Question

Judge: "What happens if a contractor fakes the GPS coordinates in a photo before uploading?"

Core Strategy: Explain multi-vector redundancy: text spoofing vs. perceptual hash of the visual scene.

Your 30-Second Spoken Answer: "That is why CHEIRAP couples GPS with 64-bit perceptual hashing. GPS is only Vector 1. Even if a contractor hacks the EXIF text to say 'Lamphelpat', our perceptual hash compares the actual visual scene against 11,202 historical projects. You can edit text coordinates in seconds, but you cannot fake the background pixels. That is how Case A immediately caught the 2024 Bishnupur classroom image."

GOVERNMENT RELEVANCE: MANIPUR OPERATIONAL REALISM (30% WEIGHTAGE) — Anticipated Jury Question

Judge: "How does this operate in remote hill areas of Manipur where mobile connectivity is zero?"

Core Strategy: Offline-first field capture with satellite GPS timestamping + asynchronous sync.

Your 30-Second Spoken Answer: "Field engineers take photos offline using standard phone cameras. Device hardware satellites record coordinates and timestamps into the EXIF layer without cellular data. When the engineer reaches the block office or Wi-Fi, the files batch-upload to CHEIRAP for hashing. Field operations never stall because of connectivity."

ADMINISTRATIVE LAW: NATURAL JUSTICE & DUE PROCESS (30% WEIGHTAGE) — Anticipated Jury Question

Judge: "Why issue a PWD Form 44 notice instead of immediately blacklisting the contractor?"

Core Strategy: Administrative due process, doctrine of natural justice, preventing High Court stays.

Your 30-Second Spoken Answer: "If an algorithm unilaterally blacklists a contractor without a hearing, the High Court stays the order within 24 hours under the doctrine of natural justice. CHEIRAP halts financial disbursements while issuing a PWD Form 44 notice with a 7-day field verification window. This upholds due process, prevents fund leakage, and gives engineers legal backing to conduct on-site audits."

ON-SITE REGISTRATION (MORNING OF 9 OCT 2026): Registration begins at 9:00 AM sharp. Arrive early, confirm HDMI/projector connection, test local servers (localhost:5173 & localhost:8000), and keep this script handy for defense.